

Antonino Spadafora

Thermal & CFD Engineer — Heat Transfer, Simulation, Thermal Management

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SUMMARY

Aerospace engineer (MSc, Politecnico di Torino) specialising in heat transfer and CFD, with 2+ years of hands-on thermal and thermo-mechanical simulation. At MBDA Italia I own the thermal characterisation of hypersonic vehicles — surface heat flux and wall temperature under extreme heating — and built a fast predictive model that puts thermal answers into the early design loop in seconds instead of hours of CFD. Comfortable across the full thermal workflow (CAD, meshing, conjugate heat transfer, thermo-mechanical FEA, validation, material trade-offs) and used to turning simulation into hardware design decisions. Seeking thermal, cooling or CFD engineering roles in hardware R&D.

EXPERIENCE

MBDA Italia — Thermal & Aerothermodynamics Engineer (MSc Thesis)

Rome · Jul 2026 – Jan 2027

- Own the thermal assessment of hypersonic vehicle geometries: surface heat flux and wall temperature prediction under extreme convective heating, driving thermal-protection sizing and material selection
- Built a reduced-order thermal model (MATLAB) that returns full heat-flux and wall-temperature fields in seconds where high-fidelity CFD takes hours, enabling thermal trade studies at the concept stage instead of after the design freezes
- Closed the model with real-gas chemical equilibrium over the whole surface — five-species equilibrium air, Blottner viscosity, Wilke's mixing rule — not only in the hottest regions
- Computed stagnation-point heat flux from Fay–Riddell, then heat-flux and wall-temperature fields over the full geometry
- Set up and ran the Ansys Fluent CFD reference cases end to end — CAD cleanup, meshing, high-speed compressible setup, convergence — and validated the model against them to quantify accuracy and define its validity envelope; findings presented to design and materials stakeholders

PoliTo Rocket Team — Politecnico di Torino

Turin · Oct 2023 – Jun 2026

Propulsion & Thermal Engineer

Sep 2024 – Jun 2026

- Ran CFD (Star-CCM+) of injector–chamber coupling to characterise mixing and internal thermal loads on a liquid rocket engine, then coupled the resulting thermal field into Ansys Mechanical thermo-mechanical FEA of the combustion chamber
- Used the coupled thermal–structural results to select chamber materials and size flanged joints against combined pressure and thermal loading; designed the injection head and injector array in SOLIDWORKS from the resulting requirements, on a hard competition deadline
- Coordinated a sub-team of 3–4 engineers, trained new members on the CFD/FEA/CAD toolchain, and acted as propulsion's interface with structures, avionics and testing

FEM Analyst

Oct 2023 – Sep 2024

- Linear and nonlinear FEA on load-bearing components and composite joints; parametric material-selection and buckling studies on a CFRP body tube under combined axial and lateral loading

TECHNICAL SKILLS

Thermal	Heat transfer (conduction, convection, radiation), conjugate heat transfer, thermal modelling, thermo-mechanical coupling, thermal trade studies, material selection
CFD & FEA	Ansys Fluent, Star-CCM+, Ansys Mechanical — meshing, solver setup, convergence, validation; compressible and high-speed flows, turbulence modelling
Modelling & Data	MATLAB, Python; reduced-order models, correlation-based methods, automation of parametric sweeps, 3D field-data post-processing
CAD	SOLIDWORKS (advanced), parametric 3D modelling, design for manufacturability

EDUCATION

MSc Aerospace Engineering — Aerogasdynamics

Politecnico di Torino, Turin · Sep 2024 – 2027 (expected)

Coursework: Computational Fluid Dynamics, Gasdynamics, Turbulent Flows, Experimental Aerodynamics, Aeroacoustics, Rocket Propulsion
Thesis (in preparation, MBDA Italia): “Preliminary models for the assessment of surface temperatures and heat fluxes on complex hypersonic glide configurations”

BSc Aerospace Engineering

Politecnico di Torino, Turin · Sep 2021 – Jul 2024

Thesis: “Flight envelopes and wing section analysis of a utility aircraft”

ADDITIONAL

Languages: Italian (native), English B2 (IELTS certified) · Endurance athlete: half-marathon running, swimming, canoe polo